

# Summary of the Li's Quasi-analytical Interpolation Method for American Options

Gary Pai

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The paper presents a quasi-analytical interpolation method for pricing American options under general multi-dimensional diffusion processes (e.g., Black-Scholes, Heston Stochastic Volatility).

## 0.1 Core Idea and Rationale

The American option price,  $P$ , is approximated by interpolating between the price of the corresponding European option,  $p$ , and the price of the Perpetual American option,  $P_\infty$ . The method is robust because it analytically determines the crucial interpolation parameter ( $A$ ) and the optimal exercise boundary ( $S_\tau^*$ ) by imposing the standard free-boundary conditions:

- (i) **Value Matching:**  $P(S_\tau^*, \tau, K) = \text{Payoff}(S_\tau^*)$ .
- (ii) **Smooth-Pasting:**  $\frac{\partial P}{\partial S}(S_\tau^*, \tau, K) = \frac{\partial \text{Payoff}}{\partial S}(S_\tau^*)$ .

## 1 The American Option Pricing Formula

For a put option with current stock price  $S$ , time to maturity  $\tau = T - t$ , and strike  $K$ , the price  $P(S, \tau, K)$  is defined piecewise:

- **Early Exercise Region** ( $S \leq S_\tau^*$ ): The price is the intrinsic value.

$$P(S, \tau, K) = K - S$$

- **Continuation Region** ( $S > S_\tau^*$ ): The price is given by the interpolation (Equation 21 in the paper):

$$P(S, \tau, K) = A \cdot \frac{P_\infty(S, K, \tilde{r})}{P_\infty(S_\tau^*, K, \tilde{r})} p(S, \tau, K e^{r\tau}) + \left(1 - A \cdot \frac{P_\infty(S, K, \tilde{r})}{P_\infty(S_\tau^*, K, \tilde{r})}\right) p(S, \tau, K) \quad (1)$$

## 2 Calculation of Intermediate Terms

The method requires computing several auxiliary terms before solving the system for  $A$  and  $S_\tau^*$ .

### 2.1 European Option Prices ( $p$ )

These are model-specific, calculated using the standard European option formula for the underlying process (e.g., Black-Scholes, Heston semi-analytical solution).

- $p(S, \tau, K)$ : Standard European put option price with strike  $K$ .
- $p(S, \tau, K e^{r\tau})$ : European put option price with a modified strike  $K' = K e^{r\tau}$ .
- $\Delta_p(S, \tau, K)$ : The Delta of the standard European put option:  $\frac{\partial p}{\partial S}(S, \tau, K)$ .

## 2.2 Modified Interest Rate ( $\tilde{r}$ ) and Time Decay Function ( $\Phi(\tau)$ )

These are used to transform the finite-maturity problem into a perpetual one for the interpolation.

- **Time Decay Function  $\Phi(\tau)$  (Equation 13):**

$$\Phi(\tau) \approx \frac{r - \delta}{r} \left( 1 - e^{-(r-\delta)\tau} \right)$$

(Where  $\delta$  is the continuous dividend yield; if  $\delta = 0$ , it simplifies to  $\Phi(\tau) \approx 1 - e^{-r\tau}$ .)

- **Modified Interest Rate  $\tilde{r}$  (Equation 15):**

$$\tilde{r} = \frac{r}{\Phi(\tau)}$$

## 2.3 Perpetual American Option Price ( $P_\infty$ )

The price  $P_\infty(S, K, \tilde{r})$  is the value of the American option at infinite maturity, solved using the modified interest rate  $\tilde{r}$ . This requires solving the time-independent PDE  $\mathcal{A}P_\infty - \tilde{r}P_\infty = 0$ .

- **General Form (Continuation Region  $S > S_p^*$ ):**

$$P_\infty(S, K, \tilde{r}) = C(\mathbf{X}) \cdot S^{q(\mathbf{X}, \tilde{r})}$$

Where  $\mathbf{X}$  represents other state variables,  $q$  is the negative root of the characteristic equation, and  $C$  is a constant determined by the perpetual critical boundary  $S_p^*$ .

- **Elasticity Term:** The elasticity  $\frac{\partial \log P_\infty}{\partial S}$  is required for  $A$ :

$$\frac{\partial \log P_\infty}{\partial S}(S, K, \tilde{r}) = \frac{1}{P_\infty(S, K, \tilde{r})} \frac{\partial P_\infty}{\partial S}(S, K, \tilde{r})$$

## 3 Solving for $A$ and $S_\tau^*$

The two unknowns are found by substituting the expression for  $A$  (from Smooth-Pasting) into the Value Matching condition, yielding a single non-linear equation for  $S_\tau^*$ .

### 3.1 Solving for $A$ as a function of $S_\tau^*$ (Smooth-Pasting)

The smooth-pasting condition ( $\frac{\partial P}{\partial S} = -1$ ) yields the interpolation parameter  $A$  (Equation 22):

$$A(S_\tau^*) = \frac{1 + \Delta_p(S_\tau^*, \tau, K)}{-D(S_\tau^*, \tau, K) \cdot \frac{\partial \log P_\infty}{\partial S}(S_\tau^*, K, \tilde{r}) - \frac{\partial D}{\partial S}(S_\tau^*, \tau, K)} \quad (2)$$

Where the difference term  $D$  and its derivative are:

$$D(S, \tau, K) = p(S, \tau, Ke^{r\tau}) - p(S, \tau, K)$$

$$\frac{\partial D}{\partial S}(S_\tau^*, \tau, K) = \frac{\partial p}{\partial S}(S_\tau^*, \tau, Ke^{r\tau}) - \frac{\partial p}{\partial S}(S_\tau^*, \tau, K)$$

### 3.2 Solving for $S_\tau^*$ (Value Matching)

Substituting  $A(S_\tau^*)$  into the Value Matching condition ( $P(S_\tau^*, \tau, K) = K - S_\tau^*$ ) at the boundary  $S_\tau^*$ , and noting that  $\frac{P_\infty(S_\tau^*, K, \tilde{r})}{P_\infty(S_\tau^*, K, \tilde{r})} = 1$ , yields the final non-linear equation for  $S_\tau^*$  (Equation 23):

$$K - S_\tau^* = A(S_\tau^*) \cdot p(S_\tau^*, \tau, Ke^{r\tau}) + (1 - A(S_\tau^*)) \cdot p(S_\tau^*, \tau, K) \quad (3)$$

### 3.3 Implementation

1. **Root Finding:** Solve Equation (3) numerically for  $S_\tau^*$ .
2. **Interpolation Parameter:** Calculate  $A$  using  $S_\tau^*$  in Equation (2).
3. **Final Price:** Calculate  $P(S, \tau, K)$  using Equation (1).